

Demo Abstract: SEQUIN: A Network Science and Physics-based Approach to Identify Sequential N-k Attacks in Electric Power Grids

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Abstract

Electric grid infrastructures are critical systems that provide power to communities, but face issues of reliability and resilience due to extreme events, e.g., natural disasters and man-made attacks. In general, the performance of the grid depends on a few key components. However, efficiently identifying them can be challenging. This demo paper presents the SEQUIN toolkit for exploring the evolution of the grid under the *sequential* failure of components. Our tool uses the specification of an electric power grid network, and provides an interface on which domain experts and practitioners can explore hypothetical attacks and their impacts. The underlying logic in the SEQUIN tool relies on network science-based principles and physics-based optimization. We envision that this tool can be used to help reason about the impact of such failures for the grid.

Keywords

resilience, power grid, N-k interdiction, sequential failure

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1 Introduction

The electric grid is a complex network of generators, buses, substations, transmission and distribution lines, which supply power to our cities and communities. Disruptions to its service from extreme events such as natural disasters and man-made attacks, e.g., Hurricane Harvey [6] and the Moore County substation terror attack [4], can have devastating consequences. In general, many grid systems rely on a few key components (generators, lines) to serve consumers, but the efficient identification of such elements, in particular combinations, is challenging. This is due to factors such as

the large, geo-distributed scale of the grid, and the automated response of the network based on physics and engineering which may potentially lead to undesirable power flows (e.g., blackout). Efforts in analyzing the performance of the grid have typically assumed that such key components fail simultaneously (or near simultaneously), and focus on the identification of these elements. However, an important observation is that the timing of such failures, i.e., *when* failures occur, is just as crucial as *where* they occur.

This paper presents our toolkit entitled SEQUIN [1] which aims to study the problem of *sequential* failures in electric power grid networks, i.e., where grid components fail one at a time. We explore this setting through a *sequential N-k interdiction model* under which two players - an attacker and a defender - interact in lockstep to produce or protect against service disruptions to the grid, respectively. In this context, our toolkit facilitates analysis of the grid from the perspective of the sequential attacker through a dashboard interface; the defender's response to attacks are done automatically as a steady state model constrained by the physics of power flows.

2 Demo

In this section, we will provide an overview of the demo for the SEQUIN toolkit. This will consist of a demo description, the user experience, and the requirements for running the demo.

2.1 Demo Description: Design of SEQUIN

This demo presents the SEQUIN toolkit, a network science and optimization based approach to construct sequential attacks on the grid, and analyze their impact. The intuition of SEQUIN lies in how the stability and adaptability of the grid under failure can be captured by structural and behavioral network properties. We employ efficient state-of-the-art physics-based models for power flow to obtain approximations of the grid dynamics. Fig. 1 depicts the SEQUIN system, which takes a power grid case (generators, buses, lines, loads, etc.), and analyzes the disruption that caused by failures. The technical details of SEQUIN are presented in [1].

The core functions of the SEQUIN system are split into four logical modules. First, in the *Network* module, the user selects the power grid network to analyze, and manages its layout in the tool. The *Attack* module then provides the guided exploration of sequential attacks on the grid, which are constructed algorithmically, e.g., using the SEQUIN approach in [1], and fine-tuned manually. This attack can then be played out using the *Visual* module, which

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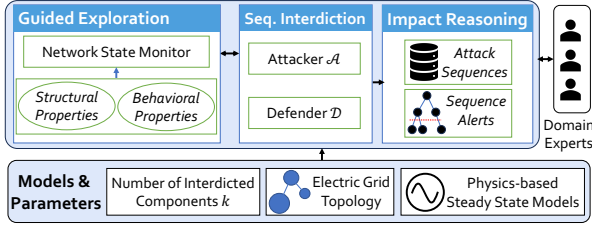


Figure 1: SEQUIN Architecture

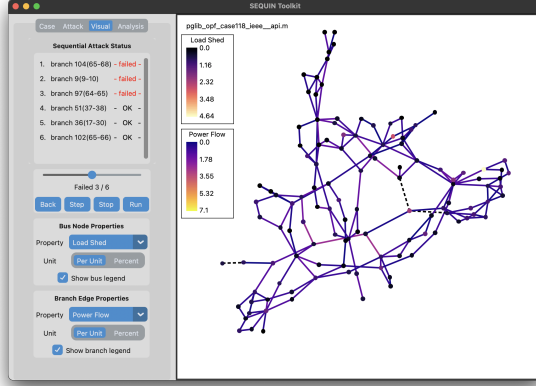


Figure 2: Screenshot of the SEQUIN Toolkit

shows the state of the network as the sequential failures occur. The *Analysis* module then displays plots showing the evolution of the state of the network. Fig. 2 depicts the SEQUIN toolkit in use.

2.2 User Experience

We next describe the user interaction with the SEQUIN toolkit.

Network Module. The user first provides an electric grid case study formatted in the Matpower specification [2], a standardized data format for power system models. This provides details about the network topology and physical attributes of its components. SEQUIN then constructs a physics-based optimization model that computes power flows and generator setpoints for the grid. Fig. 2 shows the SEQUIN toolkit on with the IEEE Case 118 benchmark (118 buses, 186 lines). The network layout and other customization options are also provided to enhance visibility, especially with larger networks. Note that the PGLib-OPF library [5] contains many benchmark case studies compatible with the SEQUIN toolkit.

Attack Module. Our toolkit provides support for many attack generation strategies, including the SEQUIN approach [1] and other heuristic methods. For a given k (the total number of failures in the grid), this provides an initial sequential attack. This demo will show the evolution of the grid as sequential failures occur, and visualize the change in certain network properties as a result of the failures. By monitoring these properties and the state of the network (consisting of power flows and generator setpoints), we will identify high-usage grid components that could lead to large disruptions; this provides insight into the grid vulnerabilities. To explore other sequences, the SEQUIN toolkit also provides a manual failure menu where users can refine attacks. As failures occur, our tool constructs and solves the appropriate physics-based optimization model. This is then visualized before the user selects the next component to

fail. For comparison, the state of the network under a simultaneous attack setting is also provided to study differences in grid responses.

Visual Module. After the user finishes selecting the branches to sequentially fail in the network, the SEQUIN toolkit can visualize the evolving state of the network as the attack progresses, as seen in Fig. 2. Here, users can select the network properties that they wish to monitor for both the bus nodes and branch edges. Ultimately, we envision that this module can be leveraged together with the Attack module to discover the conditions under which a particular network may become vulnerable, e.g., through a visualization of the state of the network before/after an attack occurs.

Analysis Module. The last module in the SEQUIN toolkit aims to generate useful plots that monitor the values of different network properties and the load shed in the network as failures occur. This includes how the degree of disruption (measured through load shed) changes across each sequential failure, and the increases/decreases in the monitored network properties. This can be used to gain useful insight into how certain failures affect the state of the network.

2.3 Demo Requirements

The SEQUIN toolkit is a software application; its dependencies are included in a public docker image. This project uses the commercial solver, Gurobi, which requires a Web License Service (WLS) license to run with docker. Detailed installation instructions for the SEQUIN toolkit are described on GitHub [3]. Our demo will use an accompanying poster to describe the electric grid use case and the key research problems addressed.

3 Conclusions and Future Work

This paper introduced the SEQUIN toolkit, which was developed to aid users in analyzing the vulnerabilities of the grid to sequential failures. In this setting, the SEQUIN architecture is implemented as four modules, which control the network ingest, sequential attack generation, visualization, and analysis. In future work, we will expand the type of attacks supported in our toolkit to include cascading failures. We also plan to incorporate more finer-grain control over other potential grid responses, e.g., load shed at nodes.

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